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| **Status** | Finished |
| **Started** | Friday, 16 October 2026, 10:14 PM |
| **Completed** | Friday, 16 October 2026, 10:18 PM |
| **Duration** | 3 mins 14 secs |
| **Grade** | **15.00** out of 15.00 (**100**%) |

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Bubble Sort
- c. Linear Search
- d. Quick Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Quick Sort
- c. Heap Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Inserting data into a list
- b. Deleting data from a list
- c. Arranging data in a particular format
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Merge Sort
- b. Quick Sort
- c. Insertion Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Greedy method
- b. Backtracking
- c. Divide and conquer
- d. Dynamic programming

The correct answer is:
Divide and conquer

Question **6**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Sorting
- b. Merging
- c. Complexity
- d. Searching

The correct answer is:
Complexity

Question **7**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By using the bubble-up method
- b. By comparing elements sequentially
- c. By breaking the input into smaller parts and merging them
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Binary Search
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Quick sort
- c. Merge sort
- d. Selection sort

The correct answer is:
Merge sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It does not require recursion
c. It sorts data using a single pass
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **11**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It recursively sorts subproblems
c. It uses the heap data structure
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **12**

Complete

Mark 1.00 out of 1.00

**The process of placing or rearranging a collection of elements into a particular order is known as**

a. Searching
b. Sorting
c. Rearranging
d. Merging

The correct answer is: Sorting

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It builds a sorted array one element at a time
- c. It compares adjacent elements and swaps them if necessary
- d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **14**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. sort()
- c. arrange()
- d. order()

The correct answer is: sorted()

Question **15**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It has no effect on the search operation
- b. It allows for faster searching
- c. It makes the search operation slower
- d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching